THE SAUNDERSON SUB-FAMILY OF THE PERFECT-CUBOID PROBLEM REDUCES TO THE RANK-ONE ELLIPTIC CURVE 160A2: AN EXPOSITORY NOTE

CΛ / LIGHTMAN CHANG

Abstract. The perfect-cuboid problem asks whether a rectangular box can have all three edges, all three face diagonals, and the space diagonal of integer length; it has been open since Euler. Two reductions of the problem to arithmetic geometry are part of the folklore: an elementary recasting in terms of three Pythagorean triples, and reductions of one-parameter cuboid sub-loci to rational points on elliptic curves, as developed by van Luijk, Sharipov, Naskręcki and most recently Peschmann. This note does not claim a new method; its purpose is to make one such reduction completely explicit on a specific sub-locus. We record the elementary equivalence between the existence of a perfect cuboid and a single quadratic relation among three Pythagorean leg-to-hypotenuse ratios, and we carry out, with full symbolic verification, the reduction of the Saunderson primitive sub-family to a square condition on the rational points of one named curve. Our only contribution is computational identification: the relevant Jacobian is Cremona curve 160a2, $y^2 = x^3 + x^2 - x + 15$, of conductor 160, rank 1 and torsion $\mathbb{Z}/2\mathbb{Z}$, with Mordell–Weil generator $(-1, 4)$; we exhibit the explicit palindromic identity realizing this and verify the absence of non-degenerate lifts in a finite range. The scope is deliberately narrow: the note settles no case left open by the cited works and is offered as a reference entry rather than a research advance.

1. Introduction

A *perfect cuboid* (or perfect Euler brick) is a rectangular box with edges a, b, c such that the three face diagonals $\sqrt{a^2+b^2}$, $\sqrt{b^2+c^2}$, $\sqrt{a^2+c^2}$ and the space diagonal $\sqrt{a^2+b^2+c^2}$ are all rational. Whether one exists is a classical open problem, recorded as Problem D18 in Guy [1], and traced to Euler. No example is known and non-existence is unproven.

Two reformulations of the problem are part of the working folklore of the subject. The first is purely elementary: a perfect cuboid is exactly a vertex at which three Pythagorean faces meet compatibly, so the problem can be phrased as a single quadratic relation among three Pythagorean triples. The second is geometric: van Luijk [3] showed that the projective surface parametrizing perfect cuboids is of general type, and various authors have studied one-parameter cuboid sub-loci by reducing them to rational points on elliptic curves; see Sharipov [5], the Mordell–Weil-rank analysis of the associated families by Naskręcki [4], and the recent quartic-reduction framework of Peschmann [6, 7]. Yoshida [8] studies the companion family of *face* cuboids in the opposite direction, constructing infinitely many of them from rational points on the same kind of curve.

The present note makes no claim to a new method. Its aim is narrow and expository: to pin down, on one explicit cuboid sub-locus, exactly *which* elliptic curve governs the reduction, in the precise sense of a Cremona label. We treat the *Saunderson primitive sub-family*, the locus of cuboids whose edges arise from the classical Saunderson parametrization (Section 3), and we show by an explicit palindromic identity that its squareness condition is a rational-point question on the single curve

$$E_{\mathrm{PCP}}: \quad y^2 = x^3 + x^2 - x + 15, \tag{1}$$

Date: May 26, 2026.

2020 *Mathematics Subject Classification.* Primary 11G05; Secondary 11D09, 11D25, 11G07.

Key words and phrases. Perfect cuboid, Euler brick, Pythagorean triple, elliptic curve, Cremona label, rational point, reformulation.

whose Cremona label is 160a2. All algebraic identities are verified symbolically and all arithmetic invariants are computed in PARI/GP; the scripts and their output are included.

Contribution and scope. The equivalence of Section 2 is folklore and is recalled, with attribution, for completeness; we claim no novelty for it. The reduction template of Section 3 is standard and is the same template used in the cited works; the curve there is a *specialization* of the kind of family those authors treat. Our single new item is the explicit identification (1) together with its invariants, stated as Theorem 1. The note settles no problem left open by [3, 5, 4, 6, 7, 8], and we recommend that the material be read as a reference entry; it is naturally a section of a larger experimental survey rather than a self-standing advance.

2. An elementary equivalence

We first record the elementary reformulation. Throughout, a *Pythagorean triple* is a triple of positive integers (p, q, r) with $p^2 + q^2 = r^2$, and we call $(p/r)^2$ a *Pythagorean square-ratio.*

Proposition 1 (Folklore; cf. [1, 6]). *There is a perfect cuboid with positive integer edges if and only if there exist three Pythagorean square-ratios P_4, P_5, P_6 with*

$$P_4 + P_6 = P_5. \tag{2}$$

Proof. Scale a putative cuboid so that the space diagonal is 1, and write $x = a/g$, $y = b/g$, $z = c/g$ with $x^2 + y^2 + z^2 = 1$. The three face conditions are that $1 - x^2$, $1 - y^2$, $1 - z^2$ be rational squares, equivalently that each of x, y, z be a rational point of the unit circle of the form $(1 - t^2)/(1 + t^2)$, i.e. a Pythagorean leg-to-hypotenuse ratio. Writing $x = \cos\theta_5$, $z = \sin\theta_4$, $y = \sin\theta_6$ with each of $\theta_4, \theta_5, \theta_6$ a rational angle, the normalization $x^2 + y^2 + z^2 = 1$ reads

$$\cos^2\theta_5 + \sin^2\theta_6 + \sin^2\theta_4 = 1, \qquad \text{i.e.} \qquad \sin^2\theta_4 + \sin^2\theta_6 = \sin^2\theta_5,$$

which is (2) with $P_i = \sin^2\theta_i$ a Pythagorean square-ratio. Conversely, three Pythagorean square-ratios satisfying (2) recover (x, y, z) and hence, after clearing denominators, integer edges (a, b, c) whose faces and space diagonal are all rational. Non-degeneracy ($abc \neq 0$) corresponds to all three ratios lying strictly between 0 and 1. $\square$

The reduction in Proposition 1 is a direct restatement of the four defining equations and is well known; the shared-leg form of the same statement appears, for instance, in the abstract and Remark 2.3 of Peschmann [6]. We include it only to fix notation and because the Saunderson reduction below is most transparent in these coordinates.

3. The Saunderson sub-family and its governing curve

We now restrict to a one-parameter sub-locus and identify its governing curve.

Definition 1. For a primitive Pythagorean triple $(p^2 - q^2,\, 2pq,\, p^2 + q^2)$, the associated *Saunderson configuration* is the cuboid sub-locus obtained by feeding this triple through the standard Saunderson primitive-quadruple parametrization

$$(p', q', n, m) = \bigl(2\lambda\mu,\, 2\lambda\nu,\, \lambda^2 - \mu^2 - \nu^2,\, \lambda^2 + \mu^2 + \nu^2\bigr),$$

and imposing the space-diagonal condition $a^2 + b^2 + c^2 = g^2$ on the resulting edges.

Eliminating the auxiliary parameters reduces the space-diagonal condition to a single squareness condition in one variable. The following identities, all verified symbolically (script `theoremE_identity.py`), make this explicit.

Lemma 1. *With $r = \mu/\nu$, the discriminant of the resulting quadratic in $s = \lambda^4/\nu^4$ is*

$$D(r) = 256\, r^2(r^2-1)^2 + 4(r^2+1)^4 = 4\big(r^8 + 68r^6 - 122r^4 + 68r^2 + 1\big). \tag{3}$$

The palindrome $r^8 + 68r^6 - 122r^4 + 68r^2 + 1$ divided by r^4 equals $W^4 + 64W^2 - 256$ under the substitution $W = r + 1/r$. Consequently the Saunderson space-diagonal condition holds for a non-degenerate configuration if and only if there is a rational number W such that

$$S^2 = W^4 + 64W^2 - 256 \tag{4}$$

for some rational S, with $W = r + 1/r$ for some rational r; the latter constraint is equivalent to $W^2 - 4$ being a non-zero rational square.

Proof. The identity (3) is a polynomial identity in $\mathbb{Z}[r]$ and is checked by expansion. Writing $W = r + 1/r$, one has $r^4 + 68r^2 - 122 + 68r^{-2} + r^{-4} = W^4 + 64W^2 - 256$, again an identity, so the condition that $D(r)$ be a perfect square is (4). Finally $W = r + 1/r$ for rational r means $r^2 - Wr + 1 = 0$ has a rational root, i.e. its discriminant $W^2 - 4$ is a (necessarily non-zero, for $r \neq \pm 1$) rational square. $\square$

The smooth projective model of (4) is a curve of genus one with the obvious rational point $(W, S) = (2, 4)$, hence an elliptic curve. The following is the single new statement of this note.

Theorem 1. *The Jacobian of the genus-one curve $S^2 = W^4 + 64W^2 - 256$ in (4) is the elliptic curve $E_{\mathrm{PCP}}\colon y^2 = x^3 + x^2 - x + 15$ of (1). Its Cremona label is* 160a2*; it has conductor* 160*, discriminant* -102400*, j-invariant $-64/25$, torsion subgroup $\mathbb{Z}/2\mathbb{Z}$ generated by $(-3, 0)$, and Mordell–Weil rank* 1*, with generator $(-1, 4)$ of canonical height* $0.17934\ldots$*. The rank equals the analytic rank, which is* 1*; in particular the rank is unconditional by the theorem of Gross–Zagier and Kolyvagin* [2].

Proof. Reducing (4) to Weierstrass form via the standard transformation for a quartic with a rational point (PARI's `ellfromeqn`) yields the curve $y^2 = x^3 + 64x^2 + 1024x + 65536$, whose minimal model is $[0, 1, 0, -1, 15]$, i.e. (1); both have j-invariant $-64/25$ (script `genus3_to_epcp.gp`). The remaining invariants are computed in PARI/GP (script `epcp_curve.gp`): `ellglobalred` gives conductor 160, `ellidentify` returns Cremona 160a2, `elltors` returns $\mathbb{Z}/2\mathbb{Z}$ generated by $(-3, 0)$, and `ellrank` returns the interval $[1, 1]$, certifying rank exactly 1 with generator $(-1, 4)$; the discriminant is -102400. The analytic rank is 1 with non-vanishing first derivative (`ellanalyticrank` returns $L'(E_{\mathrm{PCP}}, 1) = 0.9777\ldots$), so the algebraic rank equals the analytic rank unconditionally by [2]. $\square$

Corollary 1. *A non-degenerate perfect cuboid in the Saunderson sub-family of Definition 1 exists if and only if there is a point $P \in E_{\mathrm{PCP}}(\mathbb{Q})$ whose x-coordinate W satisfies that $W^2 - 4$ is a non-zero rational square.*

Proof. Immediate from Lemma 1 and Theorem 1: the lift constraint $W = r + 1/r$ is the square condition on $W^2 - 4$, and W ranges over the x-coordinates of $E_{\mathrm{PCP}}(\mathbb{Q})$. $\square$

Remark 1. Corollary 1 is a clean target but does not by itself close the Saunderson sub-family: E_{PCP} has positive rank, so $E_{\mathrm{PCP}}(\mathbb{Q})$ is infinite and the square condition must be tested along the Mordell–Weil group. Enumerating $nP_0 + \varepsilon T_0$ for $|n| \leq 300$ and $\varepsilon \in \{0, 1\}$, where $P_0 = (-1, 4)$ and $T_0 = (-3, 0)$, we find no point with $W^2 - 4$ a non-zero rational square; the only points with $W^2 - 4 = 0$ are the two degenerate solutions (script `epcp_lift.gp`). This is consistent with the expectation of non-existence but is not a proof; a complete treatment would require an effective height bound, for example via primitive-divisor methods of Silverman type, which we do not develop here.

4. Relation to prior work and limitations

The reduction template of Section 3 — from a one-parameter cuboid locus, through a genus-one quartic, to a square condition on the rational points of an elliptic curve — is standard. It is the structure of Peschmann's reduction [6], whose Corollary 5.2 states the existence of a non-degenerate point on his genus-three curve C_A in terms of a rational point P on a distinguished elliptic quotient E_A with $f(P)$ a non-trivial perfect square; the difference is only that Peschmann works with a *family* E_A parametrized by the input data, whereas our Saunderson restriction produces the single fixed curve (1). The same elliptic viewpoint underlies van Luijk [3], Sharipov [5], and the rank analysis of Naskręcki [4]; Yoshida [8] uses an isomorphic family in the opposite direction to build face cuboids.

Accordingly, the only item here that is not already implicit in those works is the explicit Cremona identification 160a2 for the Saunderson fiber, together with the palindromic identity (3) that realizes it. We make no stronger claim. In particular:

- Proposition 1 is folklore and is recalled only for self-containedness;
- Corollary 1 restricts to the Saunderson sub-locus and does not address general perfect cuboids;
- Remark 1 is a finite check, not a closure of the sub-family.

The natural home for this material is a section of an experimental survey of the perfect-cuboid problem, alongside other named-curve reductions, rather than a free-standing note.

Computations

All identities of Section 3 were verified symbolically in `sympy`; all arithmetic invariants in Theorem 1 were computed in PARI/GP 2.15.4. The scripts `theoremE_identity.py`, `n4_reconcile.py`, `epcp_curve.gp`, `genus3_to_epcp.gp` and `epcp_lift.gp`, together with their output files, accompany this note.

References

[1] R. K. Guy, *Unsolved Problems in Number Theory*, 3rd ed., Problem Books in Mathematics, Springer, New York, 2004.

[2] V. A. Kolyvagin, *Finiteness of $E(\mathbb{Q})$ and the Tate–Shafarevich group for a subclass of Weil curves*, Izv. Akad. Nauk SSSR Ser. Mat. **52** (1988), no. 3, 522–540; together with B. Gross and D. Zagier, *Heegner points and derivatives of L-series*, Invent. Math. **84** (1986), no. 2, 225–320.

[3] R. van Luijk, *On perfect cuboids*, Doctoraalscriptie (Master's thesis), Universiteit Utrecht, 2000.

[4] B. Naskręcki, *Mordell–Weil ranks of families of elliptic curves associated to Pythagorean triples*, Acta Arith. **160** (2013), no. 2, 159–183; preprint arXiv:1210.6933 [math.NT].

[5] R. A. Sharipov, *On two elliptic curves associated with perfect cuboids*, preprint arXiv:1208.1227 [math.NT], 2012.

[6] R. Peschmann, *Quartic reductions and elliptic obstructions for perfect Euler bricks*, preprint arXiv:2604.09328 [math.NT], 2026.

[7] R. Peschmann, *A torsion-intersection proof of perfect-cuboid nonexistence on* 1,072 *explicit master-tuple fibers*, preprint arXiv:2604.28072 [math.NT], 2026.

[8] T. Yoshida, *The relationship between face cuboids and elliptic curves*, preprint arXiv:2407.09825 [math.NT], 2024.

Independent Researcher
Email address: `lightman.chang@gmail.com`